Null Quantifications: How can we change the scope of quantifiers?

Rules: If variable x does not appear in A (A is a logical expression bound by a quantifier whose scope does not cover x, e.g., $\forall y\, P(y)$)

Rule 1.

$$\forall x\, (A \wedge P(x)) \equiv A \wedge \forall x\, P(x) \equiv \forall x\, P(x) \wedge A \equiv \forall x\, (P(x) \wedge A)$$

Rule 2.

$$\forall x\, (A \vee P(x)) \equiv A \vee \forall x\, P(x) \equiv \forall x\, P(x) \vee A \equiv \forall x\, (P(x) \vee A)$$

Rule 3.

$$\exists x\, (A \wedge P(x)) \equiv A \wedge \exists x\, P(x) \equiv \exists x\, P(x) \wedge A \equiv \exists x\, (P(x) \wedge A)$$

Rule 4.

$$\exists x\, (A \vee P(x)) \equiv A \vee \exists x\, P(x) \equiv \exists x\, P(x) \vee A \equiv \exists x\, (P(x) \vee A)$$

Proof or rules

- Proof: Let the domain of x be $\{x_1, \ldots, x_n\}$
- Rule 1: $\forall x\ (A \wedge P(x)) \equiv A \wedge \forall x\ P(x)$

$\forall x\ (A \wedge P(x)) \equiv (A \wedge P(x_1)) \wedge \cdots \wedge (A \wedge P(x_n)) \equiv$
$A \wedge P(x_1) \wedge \cdots \wedge P(x_n) \equiv A \wedge \forall x\ P(x)$

- Rule 2: $\forall x\ (A \vee P(x)) \equiv A \vee \forall x\ P(x)$

$\forall x\ (A \vee P(x)) \equiv (A \vee P(x_1)) \wedge \cdots \wedge (A \vee P(x_n)) \equiv$
$A \vee (P(x_1) \wedge \cdots \wedge P(x_n)) \equiv A \vee \forall x\ P(x)$

- Rule 3: $\exists x\ (A \wedge P(x)) \equiv A \wedge \exists x\ P(x)$

$\exists x\ (A \wedge P(x)) \equiv (A \wedge P(x_1)) \vee \cdots \vee (A \wedge P(x_n)) \equiv$
$A \wedge (P(x_1) \vee \cdots \vee P(x_n)) \equiv A \wedge \exists x\ P(x)$

- Rule 4: $\exists x\ (A \vee P(x)) \equiv A \vee \exists x\ P(x)$

$\exists x\ (A \vee P(x)) \equiv (A \vee P(x_1)) \vee \cdots \vee (A \vee P(x_n)) \equiv$
$(A \vee ((P(x_1) \vee \cdots \vee P(x_n)) \equiv A \vee \exists x\ P(x)$

- Note that all rules hold also for $\neg P(x)$
  - E.g., Rule 1: $\forall x\ (A \wedge \neg P(x)) \equiv A \wedge \forall x\ \neg P(x)$

PNF: Prenex Normal Form

- PNF means that all the quantifiers are in the beginning
- Every formula in predicate calculus can be written in PNF
- When I convert to PNF, I can use Rules 1-4, provided that all quantifiers are not negated, and all operators are ∨ or ∧.

Guidelines

- Use different variable names
  - ∃x P(x) → ∀**x** Q(**x**) should change to ∃x P(x) → ∀**y** Q(**y**)
- If there are operators other than ∨ or ∧, replace them with equivalent expressions that contain only ∨ or ∧
  - ∃x P(x) → ∀y Q(y) should change to ¬∃x P(x) ∨ ∀**y** Q(**y**)
- Make sure that none of the quantifiers is negated
  - ¬∃x P(x) ∨ ∀**y** Q(**y**) should change to ∀x ¬P(x) ∨ ∀**y** Q(**y**)
- Now you are ready to apply Rules 1-4
  - If you extend the scope of ∀x, and then ∀y you get ∀x∀y (¬P(x) ∨ ∀y Q(y))
  - If you extend the scope of ∀y, and then ∀x you get ∀y∀x (¬P(x) ∨ ∀y Q(y))
- ∃x P(x) → ∀**y** Q(**y**) ≡ ∀x∀y (¬P(x) ∨ ∀y Q(y)) ≡ ∀x∀y (P(x) → ∀y Q(y))

PNF Exercise

Convert $\exists z(\exists y\ R(y,z) \rightarrow \forall y\ Q(y,z))$ to PNF.

$\exists z(\exists y\ R(y,z) \rightarrow \forall y\ Q(y,z)) \equiv$ variable change from y to x

$\exists z(\exists y\ R(y,z) \rightarrow \forall x\ Q(x,z)) \equiv$ implication equivalence

$\exists z(\neg\exists y\ R(y,z) \vee \forall x\ Q(x,z)) \equiv$ De Morgan

$\exists z(\forall y\ \neg R(y,z) \vee \forall x\ Q(x,z)) \equiv$ Rule 2 on scope of $\forall x$ (A is $\forall y\ \neg R(y,z)$)

$\exists z\forall x(\forall y\ \neg R(y,z) \vee Q(x,z)) \equiv$ Rule 2 on scope of $\forall y$ (A is $Q(x,z)$)

$\exists z\forall x\forall y(\neg R(y,z) \vee Q(x,z)) \equiv$ already PNF, implication equivalence

$\exists z\forall x\forall y(R(y,z) \rightarrow Q(x,z))$

- I can always reorder quantifiers of the same type
- $\exists z\forall x\forall y(R(y,z) \rightarrow Q(x,z)) \equiv \exists z\forall y\forall x(R(y,z) \rightarrow Q(x,z))$
  - In this example it depends on which scope, $\forall x$ or $\forall y$, I extend first

- I cannot always reorder quantifiers of different type.

PNF: A Counter-intuitive Equivalence

- Recall that $\forall x\,(P(x) \vee Q(x)) \not\equiv \forall x\,P(x) \vee \forall x\,Q(x)$
  - We cannot distribute a universal quantifier over a disjunction
- Lets generate the PNF of the RHS

$\forall x\,P(x) \vee \forall x\,Q(x) \equiv$ change variable of Q

$\forall x\,P(x) \vee \forall y\,Q(y) \equiv$ apply rule 2 to change scope of x

$\forall x\,(P(x) \vee \forall y\,Q(y)) \equiv$ apply rule 2 to change scope of y

$\forall x \forall y\,(P(x) \vee Q(y))$

- Thus, $\forall x\,P(x) \vee \forall x\,Q(x) \equiv \forall x \forall y\,(P(x) \vee Q(y))$
- $\forall x \forall y\,(P(x) \vee Q(y))$ NOT EQUIVALENT to $\forall x\,(P(x) \vee Q(x))$
- The domain contains only two elements a and b such that $P(a)$ is T, $Q(a)$ is F, $P(b)$ is F, $Q(b)$ is T.
- $\forall x\,(P(x) \vee Q(x))$ is True because both $P(a) \vee Q(a)$ and $P(b) \vee Q(b)$ are T.
- $\forall x \forall y\,(P(x) \vee Q(y))$ is False because $P(b) \vee Q(a)$ is F.

PNF: Example 1 of non equivalence

- $\exists x\ P(x) \rightarrow \exists y\ Q(y)$ NOT EQUIVALENT to
- $\exists x \exists y\ (P(x) \rightarrow\ Q(y))$
- $\exists x\ P(x) \rightarrow \exists y\ Q(y) \equiv \neg\ \exists x\ P(x) \vee \exists y\ Q(y)$

Consider the domains:

Both $\neg\ \exists x\ P(x)$ and $\exists y\ Q(y)$ are false. Therefore,
$\exists x P(x) \rightarrow \exists y\ Q(y) \equiv \neg \exists x P(x) \vee \exists y\ Q(y) \equiv$ F

Domain	a	b
P	T	F
Q	F	F

- $\exists x \exists y\ (P(x) \rightarrow\ Q(y)) \equiv \exists x \exists y\ (\neg\ P(x) \vee\ Q(y))$

Using the above domains we have True for
pairs $(P(b),Q(a))$, $(P(b),Q(b))$ because $\neg\ P(b)$ is T.

- Explanation of non equivalence using PNF of $\exists x\ P(x) \rightarrow \exists y\ Q(y)$:

$\exists x\ P(x) \rightarrow \exists y\ Q(y) \equiv \neg\ \exists x\ P(x) \vee \exists y\ Q(y) \equiv$

$\forall x\ \neg P(x) \vee \exists y\ Q(y) \equiv$ by rule 2 $\forall x\ (\neg P(x) \vee \exists y\ Q(y)) \equiv$ by rule 4

$\forall x \exists y\ (\neg P(x) \vee\ Q(y)) \equiv \forall x \exists y\ (P(x) \rightarrow\ Q(y))$

Not equivalent to $\exists x \exists y\ (P(x) \rightarrow\ Q(y))$

PNF: Example 2 of non equivalence

- Is $\forall x\ P(x) \rightarrow \exists y\ Q(y)$ EQUIVALENT to $\forall x \exists y\ (P(x) \rightarrow\ Q(y))$?
- $\forall x\ P(x) \rightarrow \exists y\ Q(y) \equiv \neg\ \forall x\ P(x) \vee \exists y\ Q(y) \equiv$ DeMorgan

  $\exists x\ \neg P(x) \vee \exists y\ Q(y)\ \equiv$ (applying Rule 4 to the scope of X)

  $\exists x\ (\neg P(x) \vee \exists y\ Q(y)) \equiv$ (applying Rule 4 to the scope of y)

  $\exists x \exists y\ (\neg P(x) \vee\ Q(y)) \equiv \exists x \exists y\ (P(x) \rightarrow\ Q(y))$
- $\forall x \exists y\ (P(x) \rightarrow\ Q(y)) \equiv \forall x \exists y\ (\neg P(x) \vee\ Q(y)) \equiv$

  (applying Rule 4 to the scope of y) $\forall x\ \neg P(x) \vee \exists y Q(y) \equiv$

  $\neg\ \exists x\ P(x) \vee \exists y Q(y) \equiv \exists x\ P(x) \rightarrow \exists y\ Q(y)$
- In the previous slide we established that $\exists x\ P(x) \rightarrow \exists y\ Q(y)$ is NOT EQUIVALENT to $\exists x \exists y\ (P(x) \rightarrow\ Q(y))$

Exercise on Propositional Logic

- Which of the following is a tautology:
  1. $p \vee (p \rightarrow q)$
  2. $p \vee (q \rightarrow p)$
- Answer 1:Only 1
- Answer 2:Only 2
- Answer 3:Both 1. and 2.
- Correct answer: Answer 1:Only 1

$p \vee (p \rightarrow q) \equiv p \vee (\neg p \vee q) \equiv (p \vee \neg p) \vee q \equiv T \vee q \equiv T$

$p \vee (q \rightarrow p) \equiv p \vee (\neg q \vee p) \equiv (p \vee p) \vee \neg p \equiv p \vee \neg q$

Exercise on Predicate Logic

- Let P(x), and Q(x) be the statements “x is a professor”, “x is ignorant”.
- We saw that "All professors are not ignorant" can be expressed as $\forall x\,(P(x) \rightarrow \neg Q(x))$.
- Which of the following statements is equivalent:
  1. $\forall x\,(\neg P(x) \vee \neg Q(x))$
  2. $\neg\exists x\,(P(x) \wedge Q(x))$
- Answer 1:Only 1
- Answer 2:Only 2
- Answer 3:Both 1. and 2.
- Correct answer: Answer 3:Both 1. and 2.
- $\forall x\,(P(x) \rightarrow \neg Q(x)) \equiv \forall x\,(\neg P(x) \vee \neg Q(x)) \equiv \forall x\,\neg(P(x) \wedge Q(x)) \equiv \neg\exists x\,(P(x) \wedge Q(x))$